

w3 equations part 1

3.2 Equations involving polynomials

2nd Degree

Quadratic formula

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

factorise

quadratic formula

complete square

$$D = b^2 - 4ac$$

↑
discriminant

$$D < 0 \rightarrow \text{no sol}$$

$$D = 0 \rightarrow 1 \text{ sol}$$

$$D > 0 \rightarrow 2 \text{ sol}$$

Complete the square

$$\underbrace{x^2 + 4x + 1 = 0}_{|} \rightarrow (x+r)^2$$

$$x^2 - 2rx + r^2$$

$$(x-r)^2 - r^2 = x^2 - 2rx$$

$$(x+2)^2 - 4 + 1 = 0$$

$$-2r = 4 \Rightarrow \boxed{r = -2}$$

$$(x+2)^2 = 3$$

$$x+2 = \sqrt{3} \quad \text{or} \quad x+2 = -\sqrt{3}$$

$$x = -2 + \sqrt{3} \quad \text{or} \quad x = -2 - \sqrt{3}$$

$$\text{Method} \rightarrow a(x-r)^2 = -s$$

$$(x-r)^2 = \frac{-s}{a}$$

$$x-r = \sqrt{\frac{-s}{a}} \quad \text{or} \quad x-r = -\sqrt{\frac{-s}{a}}$$

$$x = r + \sqrt{\frac{-s}{a}} \quad \text{or} \quad x = r - \sqrt{\frac{-s}{a}}$$

polynomial
is
continuous
 $f(x)$

Degree 3 or higher

→ Number of solutions: odd degree

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

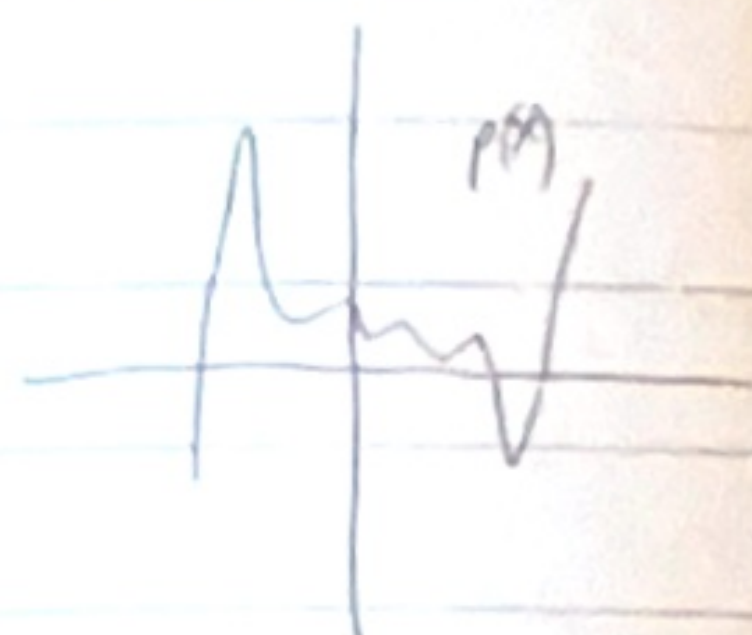
n odd

large $|x|$: $p(x) \sim a_n x^n$

if $a_n > 0$ then

$$p(x) > 0 \text{ if } x \gg 0$$

$$p(x) < 0 \text{ if } x \ll 0$$



→ Even degree

$$g(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

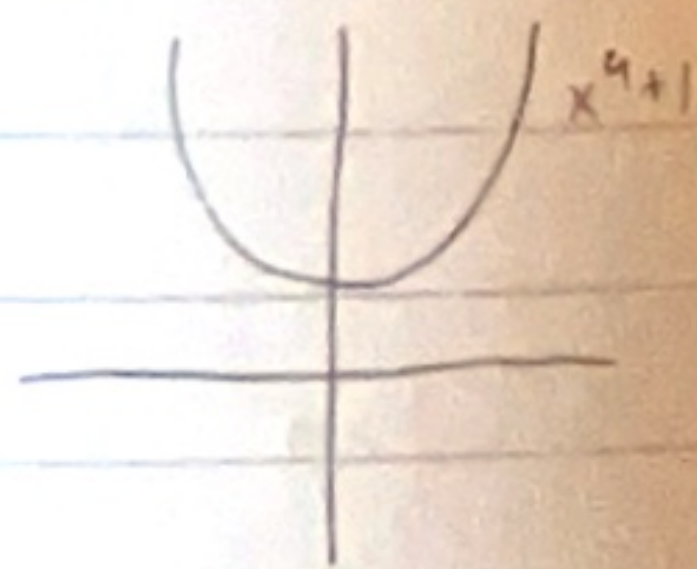
n even

example

$$x^4 + 1 = 0 \text{ has no solutions}$$

even degree

$g(x) = 0$ may have no solutions



degree 4 or higher

- factor

- factorization

- complex factor

Factorization

$$(x-1)(ax^2 + bx + c) = 1x^3 + 0x^2 + (-7)x + 6$$

$$(x-1)(ax^2 + bx + c) = ax^3 + (b-a)x^2 + (c-b)x - c$$

↓

$$\begin{array}{lcl} a=1 & \rightarrow & a=1 \\ b-a=0 & \rightarrow & b=1 \\ c-b=-7 & \rightarrow & c=-6 \end{array}$$

w4. Equations

TRIGONOMETRIC equations

A way of describing a solution

$$x = \frac{\pi}{6} + 2\pi k, \quad k \text{ integer}$$

Since $\sin$ is periodic, these are infinite solutions

Simplify to $\sin(x) = c$

$$\cos(x) = c$$

→ Use substitution

→ use trigonometric identity

EQUATIONS WITH EXPONENTIALS AND LOGARITHMS

Ty and reduce the equation to:

$$a^p = a^q$$

$\Leftrightarrow$

$$p = q$$

Two main techniques $\rightarrow$

1) $\rightarrow$ Change everything to the same technique

$$4x = 2^{x^2}$$

$$2^{2x} = 2^{x^2}$$

$$2x = x^2$$

$$x = 0$$

$$x = 2$$

2) Substitution

$$2^x + 4^x = 6$$

$\downarrow$

Change to the same base

$$2^x + 4^x = 6$$

$$2^x + 2^{2x} = 6$$

$\downarrow$

$$4 + y^2 = 6$$

$$\text{if } y = 2^x$$

$$\rightarrow \boxed{\begin{matrix} y = -3 \\ y = 2 \end{matrix}}$$

then only solution

$$\uparrow \boxed{x=1}$$

not negative

$$\boxed{2^x = -3}$$

LOGS

1) Use a single base

2) Substitution

3) Simplify

$$\log_a(p) = \log_a(q) \Leftrightarrow p = q$$

Complications

$$\ln(x) + \ln(2x-3) = \ln(2)$$

$$\ln(x(2x-3)) = \ln(2)$$

$\downarrow$

$$x(2x-3) = 2$$

$$2x^2 - 3x - 2 = 0$$

$$(2x+1)(x-2) = 0$$

$$\boxed{x = -\frac{1}{2} \text{ or } x = 2}$$

$\downarrow$

doesn't exist

$\downarrow$ no solution

then

inequalities

> <

strict

less

greater

smaller

larger

→ The boundaries of the interval are either marked by a point where both sides of the inequality are equal or a point where a domain problem occurs

example

$$x \leq \frac{4}{x}$$

solve for a problem of domain on

-2

0

2



Agree on points of each interval & substitute in the equation & see if it works

for eg $x = -4 \rightarrow -4 \leq \frac{4}{-4}$

$$-4 \leq -1$$

✓

→ work on this interval the inequality is correct

greater than less than

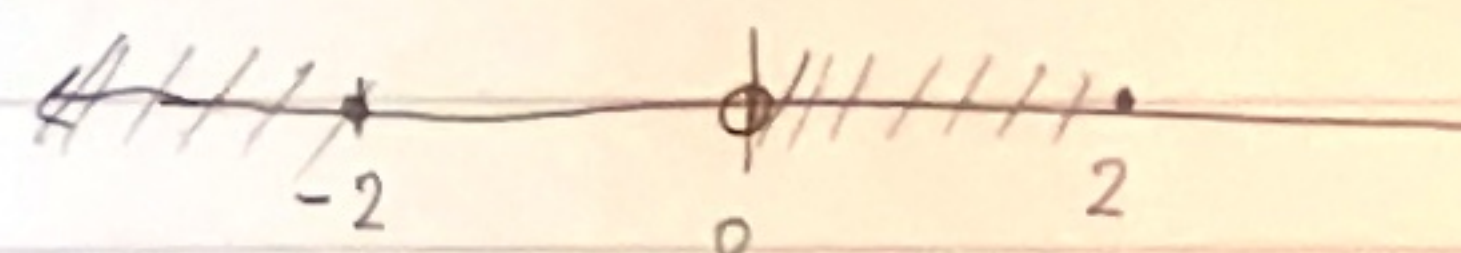
-2 ✓

+2 ✓

The inequality is satisfied for

$$x \leq -2 \text{ or}$$

$$0 < x \leq 2$$



solution set:

$$[-\infty, -2] \cup (0, 2]$$

Step 1

find points at which

both sides are equal

$f(x)$ is not defined or discontinuous

Step 2 check inequality in each interval

Step 3 check " at each boundary point